

Notes: De La Roca and Puga (REStud, 2017)

Learning by Working in Big Cities

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 Why Spain is a useful laboratory
- 2.2 Data source and panel construction
- 2.3 Sample restrictions
- 2.4 Urban areas and city-size measure
- 2.5 Key outcome and control variables
- 2.6 City-specific experience measures
- 2.7 Mobility facts that matter for identification

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Economic mechanism and hypotheses
- 3.2 Full wage equation
- 3.3 Static pooled OLS specification
- 3.4 Static worker fixed-effects specification
- 3.5 Dynamic experience specification
- 3.6 Short-term versus medium-term city premia
- 3.7 Heterogeneous dynamic benefits by worker ability
- 3.8 Why the pooled-OLS to fixed-effects drop is not “pure sorting”

4 Identification and instruments (what makes the estimates credible)

- 4.1 What identifies the baseline estimates
- 4.2 Why standard worker fixed effects are not enough
- 4.3 Why the dynamic specification is credible
- 4.4 Endogeneity of city size and the IV strategy

4.5	Balanced migration flows and why the static FE estimate works	
4.6	Ashenfelter-dip check	
4.7	Mover-stayer selection check	
4.8	Other robustness checks	
4.9	What the paper can and cannot distinguish	
4.10	Relation to the literature	
5	Tables and figures	
5.1	Figure 1: mean earnings and city size	
5.2	Table 1 and Figure 2: the static city-size premium	
5.3	Table 2 and Figures 3–5: dynamic benefits, portability, and medium-term premia . .	
5.4	Table 3 and Figure 6: IV evidence and the no-Ashenfelter-dip check	
5.5	Table 4 and Figure 7: heterogeneity by worker ability	
5.6	Table 5, Figure 8, and Table 6: what sorting really looks like	
6	Short summary	
7	Conclusion	
7.1	Core conclusion (what the paper establishes)	
7.2	Economic intuition	
7.3	Takeaway	

1 Big picture

- **Question.** Why do workers in bigger cities earn more? Is the premium mainly due to static agglomeration advantages, sorting of inherently better workers into large cities, or learning by working in big cities?
- **Setting.** The paper studies Spain using administrative data that track workers over time, across firms, and across cities. The main estimation sample focuses on male Spanish-born workers and uses uncensored earnings data for 2004–2009 together with full labor-market histories back to 1980.
- **Core challenge.** A simple wage premium for big cities is hard to interpret. It may reflect genuinely higher productivity while working there, better workers sorting into large cities, or the fact that experience accumulated in large cities becomes more valuable over time.
- **Main conceptual contribution.** The paper jointly estimates *static city effects*, *worker fixed effects*, and *city-specific returns to experience*. This lets it separate immediate gains from current location, accumulated gains from big-city learning, and sorting on permanent worker ability.
- **Main hypotheses.**
 - **Static agglomeration:** workers earn more while currently employed in larger cities.
 - **Sorting:** larger cities may attract workers with higher time-invariant ability.
 - **Dynamic learning:** experience acquired in bigger cities is more valuable and can remain valuable after workers move away.
- **Main empirical design.** The paper starts with a standard pooled OLS city-size wage premium, then adds worker fixed effects, and then estimates a richer specification in which the value of experience depends on where it was acquired and where it is used. A final extension allows the returns to big-city experience to vary with worker ability.
- **Main baseline result.** The pooled OLS elasticity of earnings with respect to city size is 0.0455. Once worker fixed effects are added, the static elasticity falls to 0.0241. In the full dynamic specification, the short-term static elasticity is 0.0223 but the *medium-term* elasticity rises to 0.0510. In the medium term, about half of the city-size premium is static and about half is dynamic.
- **Main portability result.** Experience acquired in the biggest cities is highly portable. Workers keep most of the extra value of that experience even after relocating to smaller cities.
- **Main heterogeneity result.** Big-city experience is especially valuable for high-ability workers. Thus, larger cities raise not only average earnings but also the dispersion of earnings.
- **Main sorting result.** There is clear sorting on observable occupational categories, but within these broad groups there is little additional sorting on unobserved ability. Workers in big and small cities are not very different to start with; working in cities of different sizes is what makes their earnings diverge.
- **Economic takeaway.** Bigger cities matter mainly because they provide an immediate productivity premium and a dynamic learning premium that becomes embedded in human capital. The traditional interpretation that the city-size wage premium mainly reflects sorting on unobservables is too strong.

2 Data and measurement (key objects)

2.1 Why Spain is a useful laboratory

- Spain exhibits a strong city-size earnings gradient. In the paper’s motivating example, average annual earnings of male employees are about 31,000 euros in Madrid, 21% higher than in Valencia, 46% higher than in Santiago de Compostela, and 55% higher than in rural areas.
- The country has rich administrative data that allow the authors to observe actual employment histories, firm locations, occupations, contracts, and uncensored earnings.
- Direct observation of workplace location is crucial because the paper wants to know *where experience was acquired*, not just where the worker lives.

Interpretation. Spain is valuable here because the earnings premium for large cities is economically meaningful and the data are rich enough to separate static and dynamic channels.

2.2 Data source and panel construction

- The main data set is Spain’s *Continuous Sample of Employment Histories* (MCVL), which matches social security records, income-tax records, and census information.
- The MCVL is a 4% random sample of everyone who has some relationship with the Spanish Social Security system in a given year.
- Social security records provide complete job histories since first employment, or since 1980 for earlier entrants.
- Income-tax records provide uncensored labor earnings, available yearly and then converted into monthly earnings per day of full-time-equivalent work.
- By combining MCVL editions from 2004 onward, the paper constructs a monthly panel with complete labor histories since 1980 and uncensored earnings from 2004 to 2009.

Interpretation. The data allow the authors to observe workers repeatedly over their careers and to reconstruct how much experience each worker accumulated in different places.

2.3 Sample restrictions

The baseline sample begins with men aged 18 and above, with Spanish citizenship, born in Spain since 1962, and employed at some point between January 2004 and December 2009.

Main exclusions are:

- women, because Spain’s female labor-force participation changed dramatically over the period relevant for measuring lifetime experience;
- foreign-born workers, because their pre-immigration labor histories are unobserved;
- self-employment spells, because comparable labor earnings are unavailable there;
- jobs in the Basque Country and Navarre, where comparable tax-return earnings data are unavailable;

- rural areas and three very small urban areas that cannot be identified precisely in the MCVL;
- agriculture, fishing, mining, and other extractive activities with unreliable earnings-day records;
- public-sector, education, health, and other heavily regulated sectors where pay is not market-determined in the same way.

The final sample has **157,113 workers** and **6,263,446 monthly observations**.

Interpretation. The sample is deliberately narrow. The paper wants a clean test of urban earnings premia among workers whose full employment histories can be measured.

2.4 Urban areas and city-size measure

- The paper starts from 85 official urban areas defined by Spain's Ministry of Housing.
- After excluding the four urban areas in the Basque Country and Navarre, three very small urban areas, and the enclaves of Ceuta and Melilla, the estimation sample contains **76 urban areas**.
- City size is measured as the *number of people within 10 km of the average person* in the urban area, computed from a high-resolution 1 km population grid.

This measure is preferred to simple population or density because it handles polycentric urban areas better and is less sensitive to boundary definitions.

Interpretation. The key right-hand-side variable is not raw population. It is an effective local scale measure intended to capture how much population is actually nearby.

2.5 Key outcome and control variables

The central outcome is log monthly labor earnings per day of full-time-equivalent work.

Other key controls are:

- overall labor-market experience, measured from actual days worked;
- tenure in the current establishment;
- occupation, grouped into five skill categories;
- contract type and working hours;
- two-digit sector indicators;
- month-year indicators;
- worker fixed effects and city fixed effects in the richer specifications.

Interpretation. The paper's identification does not come from comparing workers with vaguely similar resumes. It comes from tracking exactly where each worker gained experience and how wages changed after moving.

2.6 City-specific experience measures

A distinctive feature of the paper is that cumulative experience is tracked by location.

Let:

- e_{ijt} = experience worker i has accumulated in city j by time t ;
- E_{it}^{1-2} = experience in the two biggest cities (Madrid and Barcelona);
- E_{it}^{3-5} = experience in the third to fifth biggest cities (Valencia, Sevilla, Zaragoza);
- E_{it}^o = experience outside the five biggest cities;
- E_{it} = total experience.

Interpretation. This is the core measurement innovation. The same total experience can have different value depending on where it was acquired.

2.7 Mobility facts that matter for identification

- Workers are assigned to cities using the *workplace municipality*, so moves are job-location moves across urban areas.
- About **7%** of workers change jobs across urban areas each year.
- In the sample there are 40,809 cross-city migrations: 8,868 from the five biggest cities to smaller cities, 8,790 from smaller cities to the five biggest cities, and 23,151 between cities of similar sizes.

Interpretation. Balanced flows between big and small cities become important later because they help explain why static fixed-effects estimates are not badly biased by omitted dynamic learning.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper argues that the city-size wage premium may come from three sources:

1. a **static** premium from currently working in a larger city;
2. **sorting** of initially better workers into larger cities;
3. a **dynamic** premium because experience acquired in larger cities is more valuable.

The key challenge is that standard pooled OLS and standard worker fixed-effects models mix these channels together.

3.2 Full wage equation

The conceptual starting point is:

$$\log w_{ict} = \sigma_c + \mu_i + \sum_{j=1}^C \delta_{jc} e_{ijt} + x'_{it} \beta + \varepsilon_{ict}. \quad (1)$$

Here:

- σ_c is a city fixed effect,
- μ_i is a worker fixed effect,
- e_{ijt} is experience worker i has accumulated in city j by time t ,
- δ_{jc} measures the value of experience acquired in city j when it is used in city c ,
- x_{it} contains observed time-varying controls,
- ε_{ict} is the error term.

Interpretation.

- σ_c captures the *static* advantage of currently working in city c .
- μ_i captures time-invariant worker ability and therefore sorting on unobservables.
- The δ_{jc} terms capture the *dynamic* advantage of having acquired experience in different cities.

The empirical versions of the model add nonlinear terms so the value of extra experience can decline with total experience.

3.3 Static pooled OLS specification

The simplest model ignores both worker heterogeneity and city-specific experience:

$$\log w_{ict} = \sigma_c + x'_{it}\beta + \eta_{ict}. \quad (2)$$

The paper estimates this by pooled OLS and then, in a second stage, regresses the estimated city effects on log city size.

Interpretation. This is the standard “static city premium” model. It is easy to estimate but it is biased upward if larger cities contain better workers or workers with more valuable accumulated experience.

3.4 Static worker fixed-effects specification

Following Glaeser and Maré and Combes et al., the paper then adds worker fixed effects:

$$\log w_{ict} = \sigma_c + \mu_i + x'_{it}\beta + \zeta_{ict}. \quad (3)$$

After subtracting worker means, the city effects are identified from migrants:

$$(\log w_{ict} - \overline{\log w_i}) = \sum_{c=1}^C \sigma_c (\iota_{ict} - \bar{\iota}_{ic}) + (x_{it} - \bar{x}_i)' \beta + (\zeta_{ict} - \bar{\zeta}_i). \quad (4)$$

Interpretation.

- This specification removes sorting on time-invariant worker ability.
- But it still omits the dynamic value of big-city experience, so the drop from pooled OLS to worker fixed effects does *not* automatically measure sorting.

3.5 Dynamic experience specification

The main empirical model allows the value of experience to differ by where it was acquired and where it is currently used. A simplified version of the estimated specification is:

$$\begin{aligned} \log w_{ict} = & \sigma_c + \mu_i + \beta_1 E_{it}^{1-2} + \beta_2 (E_{it}^{1-2} \times E_{it}) + \beta_3 E_{it}^{3-5} + \beta_4 (E_{it}^{3-5} \times E_{it}) \\ & + \beta_5 E_{it} + \beta_6 E_{it}^2 + \theta_1 (E_{it}^{1-2} \times I_{ct}^5) + \theta_2 (E_{it}^{1-2} \times E_{it} \times I_{ct}^5) \\ & + \theta_3 (E_{it}^{3-5} \times I_{ct}^5) + \theta_4 (E_{it}^{3-5} \times E_{it} \times I_{ct}^5) \\ & + \theta_5 (E_{it}^o \times I_{ct}^5) + \theta_6 (E_{it}^o \times E_{it} \times I_{ct}^5) + x'_{it}\beta + \varepsilon_{ict}, \end{aligned} \quad (5)$$

where I_{ct}^5 is an indicator for currently working in one of the five biggest cities.

Interpretation.

- β_1 and β_3 say how much extra value is attached to experience accumulated in the biggest cities relative to experience accumulated elsewhere.
- The θ terms test portability by asking whether the value of experience changes when the worker uses it inside or outside the five biggest cities.
- If these interaction terms are close to zero, experience acquired in big cities is portable.

Empirically, the first year of experience in Madrid or Barcelona raises earnings by about 3.1% relative to a year acquired outside the top five, while a first year in a city ranked third to fifth raises earnings by about 1.5%.

3.6 Short-term versus medium-term city premia

Once the first-stage dynamic model is estimated, the paper constructs two second-stage objects.

Short-term premium.

$$\hat{\sigma}_c = a + b \log(\text{city size}_c) + u_c, \quad (6)$$

where $\hat{\sigma}_c$ is the estimated city fixed effect from the dynamic first stage. The slope b is the elasticity of the *static* city premium with respect to city size.

Medium-term premium.

$$MP_c = \hat{\sigma}_c + \hat{V}_c(7.72), \quad (7)$$

where $\hat{V}_c(7.72)$ is the estimated value of 7.72 years of experience acquired in city c , the sample average amount of experience accumulated in one location.

The second-stage regression is again:

$$MP_c = a + b^{MT} \log(\text{city size}_c) + u_c. \quad (8)$$

Interpretation.

- b measures the immediate premium from arriving in a bigger city.
- b^{MT} measures the premium after working there long enough to accumulate local experience.

In the paper, the short-term elasticity is 0.0223 and the medium-term elasticity is 0.0510.

3.7 Heterogeneous dynamic benefits by worker ability

To allow more able workers to benefit more from big-city experience, the paper extends the model to:

$$\log w_{ict} = \sigma_c + \mu_i + \sum_{j=1}^C (\delta_j + \phi_j \mu_i) e_{ijt} + x'_{it} \beta + \varepsilon_{ict}. \quad (9)$$

Here ϕ_j captures how the value of experience acquired in city j varies with worker ability μ_i .

The worker effect is then updated recursively via:

$$\hat{\mu}_i = \frac{\log w_{ict} - \hat{\sigma}_c - \sum_{j=1}^C \hat{\delta}_j e_{ijt} - x'_{it} \hat{\beta}}{1 + \sum_{j=1}^C \hat{\phi}_j e_{ijt}}. \quad (10)$$

Interpretation. If $\phi_j > 0$ for large cities, big-city learning and worker ability are complements. That means large cities widen the earnings distribution as well as raise average earnings.

3.8 Why the pooled-OLS to fixed-effects drop is not “pure sorting”

The paper’s algebra shows that once city-specific experience is omitted, both pooled OLS and worker fixed-effects estimates of the static premium are biased. With balanced flows between big and small cities and highly portable big-city experience, the bias in the worker fixed-effects estimate is likely small. But the gap between pooled OLS and worker FE is *not* automatically evidence of sorting, because dynamic learning can generate the same pattern.

Interpretation. This is one of the paper’s main conceptual points. Earlier literature tended to interpret the fall from 0.0455 to 0.0241 as sorting. De La Roca and Puga argue that learning by working in big cities can explain much of that fall.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimates

The static city effects in worker fixed-effects models are identified by workers who change cities. The experience coefficients are identified by observing the same worker accumulate experience over time, both for stayers and for movers. The dynamic model exploits the fact that the same amount of total experience can be composed of very different histories of where that experience was acquired.

Interpretation. The source of identification is not cross-sectional comparison of different workers. It is within-worker variation over time and across locations.

4.2 Why standard worker fixed effects are not enough

Worker fixed effects control for time-invariant unobserved ability, but they do not remove bias from omitted dynamic learning. A migrant moving from a small city to a big city gains both an immediate static premium and starts accumulating more valuable experience; a migrant moving from a big city to a small city may continue to benefit from previously acquired big-city experience.

Interpretation. Without modeling the dynamic component explicitly, the static premium is forced to absorb too much.

4.3 Why the dynamic specification is credible

The core credibility result is portability. Experience acquired in Madrid or Barcelona does not lose much of its value when the worker later moves to a smaller city. That is hard to reconcile with a purely local static story and is naturally consistent with learning embedded in human capital.

Interpretation. Portability is the paper's most persuasive evidence that the dynamic premium reflects actual learning rather than local amenities or contemporaneous firm composition.

4.4 Endogeneity of city size and the IV strategy

The first-stage wage regressions deal with sorting and learning, but the second-stage regression of city premia on city size could still be biased if some omitted city characteristic both raises wages and attracts people.

To address this, the paper instruments current city size with:

- city size in 1900,
- land fertility within 25 km of city center,
- percentage of water within 25 km,
- percentage of steep terrain within 25 km,
- log mean elevation within 25 km,
- Roman road rays crossing a 25 km circle around city center.

Interpretation. These variables affect long-run city size through historical population persistence, agricultural suitability, land supply, topography, and transportation history, but are plausibly exogenous to today's city wage premium except through city size.

The IV results are very close to OLS:

- short-term elasticity: 0.0203 instead of 0.0223,
- medium-term elasticity: 0.0530 instead of 0.0510.

Takeaway. Endogeneity of city size does not appear to be driving the main results.

4.5 Balanced migration flows and why the static FE estimate works

There are roughly as many migrations from the top five cities to smaller cities as in the opposite direction. If learning benefits are highly portable, these roughly offsetting flows mean that omitted dynamic benefits create little net bias in the standard fixed-effects estimate of the static premium.

Interpretation. This explains why the static elasticity from the rich dynamic model, 0.0223, is so close to the standard fixed-effects estimate, 0.0241.

4.6 Ashenfelter-dip check

A concern is that workers might move only after an adverse earnings shock, which would exaggerate the measured gain from moving to a big city. The paper addresses this by estimating event-time indicators for eight quarters before and after a migration.

Result: there is **no Ashenfelter dip** before migration. The pre-move relative earnings profile is flat, and the post-move path matches the benchmark parametric model.

Interpretation. The measured big-city premium is not an artifact of workers moving immediately after bad wage draws.

4.7 Mover-stayer selection check

The paper also allows the value of big-city experience to differ for stayers, migrants into big cities, and migrants out of big cities. Migrants do have steeper overall earnings profiles, but the extra value of experience acquired in the biggest cities is not statistically different across these groups.

Interpretation. The main learning coefficients are not being driven by unusually selected movers.

4.8 Other robustness checks

- Re-estimating the model for 1998–2003 gives a short-term elasticity of 0.0247 and a medium-term elasticity of 0.0439, close to the baseline results.
- Repeating the analysis for women yields a much smaller medium-term elasticity of 0.0229.
- Including public, health, education, and other regulated sectors lowers the medium-term elasticity from 0.0510 to 0.0431.

Interpretation. The main findings are not specific to the housing-boom years 2004–2009, but they are strongest for men and for the more market-driven private sector.

4.9 What the paper can and cannot distinguish

The paper is strong on separating static effects, dynamic learning, and time-invariant unobserved ability. It is weaker on fully specifying the general-equilibrium mechanism behind urban advantages. It also does not study welfare or spatial equilibrium directly, and the baseline sample excludes women, the self-employed, immigrants, rural jobs, and heavily regulated sectors.

Interpretation. The contribution is very sharp on the wage-measurement problem, but it is not a full urban general-equilibrium model.

4.10 Relation to the literature

The paper directly engages with Glaeser and Maré, Combes et al., Baum-Snow and Pavan, and Eeckhout et al.

Its main contribution relative to that literature is:

- to show that the fall from pooled OLS to worker fixed effects is not clean evidence of sorting;

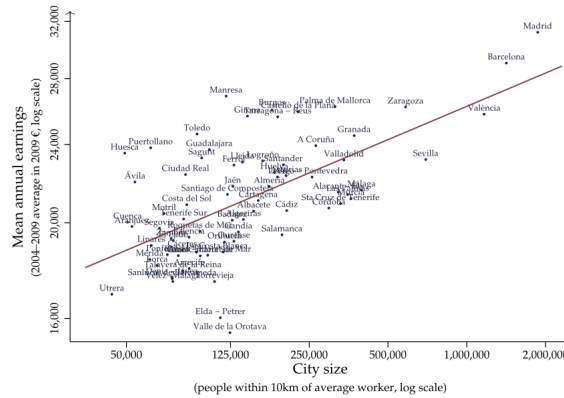


FIGURE 1
Mean earnings and city size

- to quantify the dynamic value of experience accumulated in larger cities;
- to show that greater earnings dispersion in larger cities can arise from the interaction of ability and learning, not only from stronger sorting.

5 Tables and figures

5.1 Figure 1: mean earnings and city size

Read:

- Figure 1 plots mean annual earnings against city size for Spanish urban areas.
- The relationship is clearly upward sloping.
- Madrid and Barcelona lie at the upper end of both earnings and city size.

Economic message. Before any controls, bigger cities plainly pay more. The rest of the paper asks whether this reflects static productivity advantages, sorting, or learning.

5.2 Table 1 and Figure 2: the static city-size premium

Read:

- In Table 1, the pooled OLS elasticity of the earnings premium with respect to city size is 0.0455.
- Once worker fixed effects are added, the elasticity falls to 0.0241.
- Figure 2 plots the pooled-OLS city effects against log city size and shows a clear upward relation.
- The pooled OLS regression has an R^2 of 0.2406 in the second stage, while the fixed-effects version has an R^2 of 0.1422.

Economic message. A standard reading would say that the drop from 0.0455 to 0.0241 is sorting. The paper's point is that this interpretation is incomplete because dynamic learning is omitted from both specifications.

relative to a year outside the top five; a first year in cities ranked third to fifth adds about 1.5%.

- Table 2, column (2) gives a short-term static city-size elasticity of 0.0223.
- Table 2, column (3) gives a medium-term elasticity of 0.0510, evaluated at 7.72 years of local experience.
- Figure 3 shows that relative to Santiago, working in Madrid yields about a 9% initial premium and about a 36% premium after 10 years; Sevilla gives roughly 3% initially and 14% after 10 years.
- The dashed lines in Figure 3 show that workers keep most of the gain from experience acquired in bigger cities after moving away.
- Figure 4 shows that workers with prior experience in Santiago get a higher immediate jump when they move to Madrid but a flatter subsequent profile: over 10 years the Madrid-versus-Santiago gain is about 31% for an experienced mover versus 36% for an inexperienced worker.
- Figure 5 plots the medium-term premium against city size and shows a tighter relation than the pooled OLS plot: the second-stage R^2 rises from 0.2406 to 0.3732.

TABLE 4
Estimation of the heterogeneous dynamic and static city size earnings premia

Dependent variable	(1) Log earnings	(2) Initial premium (city indicator coefficients column (1))	(3) Medium-term premium (initial + 7.7 years local experience)
Log city size		0.0243 (0.0061)***	0.0495 (0.0108)***
City indicators	Yes		
Worker fixed effects	Yes		
Experience first to second biggest cities	0.0293 (0.0022)***		
Experience first to second biggest cities × experience	-0.0007 (0.0001)***		
Experience third to fifth biggest cities	0.0143 (0.0043)***		
Experience third to fifth biggest cities × experience	-0.0006 (0.0003)**		
Experience	0.0979 (0.0007)***		
Experience ²	-0.0009 (0.0000)***		
Experience first to second biggest × worker fixed effect	0.0097 (0.0030)***		
Experience first to second biggest × experience	-0.0001 (0.0001)		
Experience third to fifth biggest cities	0.0042 (0.0045)		
Experience third to fifth biggest × experience	-0.0001 (0.0002)		
Experience × worker fixed effect	0.0632 (0.0034)***		
Experience ² × worker fixed effect	-0.0021 (0.0001)***		
Experience first to second biggest	-0.0034 (0.0022)		
Experience first to second biggest × experience	0.0001 (0.0001)		
Experience third to fifth biggest × experience	-0.0029 (0.0045)		
Experience outside five biggest × experience	0.0003 (0.0003)		
Experience outside five biggest × experience	0.0022 (0.0024)		
Experience outside five biggest × experience	0.0000 (0.0001)		
Observations	6,263,446	76	76
R^2	0.1228	0.1352	0.3439

Notes: All regressions include a constant term. Column (1) also includes firm tenure and its square, occupation indicators, month-year indicators, two-digit sector indicators, and contract-type indicators. Coefficients in column (1) are reported with bootstrapped standard errors in parenthesis which are clustered by worker (achieving convergence of coefficients and mean squared error of the estimation in each of the 100 bootstrap iterations). Coefficients in columns (2) and (3) are reported with robust standard errors in parenthesis. ***, **, and * indicate significance at the 1, 5, and 10% levels. The R^2 reported in column (1) is within workers. Worker values of experience and tenure are calculated on the basis of actual days worked and expressed in years. City medium-term premium calculated for workers' average experience in one city (7.72 years).

TABLE 5
Comparison of occupational groups across cities of different sizes

	Occupational groups (%)				
	Very-high-skilled	High-skilled	Medium-high-skilled	Medium-low-skilled	Low-skilled
First to second biggest cities	10.9	13.8	24.2	41.7	9.4
Third to fifth biggest cities	6.3	10.9	21.0	48.2	13.8
Other cities	3.5	7.9	18.4	54.0	16.1

Notes: Employers assign workers into one of ten social security categories which we regroup into five occupational skill categories. Shares are averages of monthly observations in the sample.

Economic message. The medium-term city-size premium is much larger than the static premium because bigger cities are learning environments. The portability result is central: workers keep much of what they learned.

TABLE 3
IV estimation of the dynamic city size earnings premium

Dependent variable	(1) Log size	(2) Short-term premium	(3) Medium-term premium
Instrumented log city size		0.0203 (0.0079)***	0.0530 (0.0143)***
Log city size 1900	0.6489 (0.0810)***		
High-quality land within 25 km of city centre (%)	0.0151 (0.0065)**		
Water within 25 km of city centre (%)	0.0059 (0.0029)**		
Steep terrain within 25 km of city centre (%)	-0.0134 (0.0057)**		
Log mean elevation within 25 km of city centre	0.2897 (0.0834)***		
Roman road rays 25 km from city centre	0.0674 (0.0372)*		
Observations	76	76	76
R ²	0.6503	0.1271	0.3726
F-test weak ident. (H_0 : instruments jointly insignificant)		25.2482	25.2482
P-value LM test (H_0 : model underidentified)		0.0236	0.0236
P-value J test (H_0 : instruments uncorr. with error term)		0.3025	0.2051
P-value endog. test (H_0 : exogeneity of instrumented var.)		0.5757	0.5998

Notes: All regressions include a constant term. Column (1) is the first-stage regression of log city size on a set of historical population and geographical instruments. Columns (2) and (3) are second-stage regressions of city premia on instrumented log city size. Coefficients are reported with robust standard errors in parenthesis. ***, **, and * indicate significance at the 1, 5, and 10% levels. The F-statistic (or Kleibergen-Papp rk Wald statistic) reported on the weak instruments identification test exceeds all thresholds proposed by Stock and Yogo (2005) for the maximal relative bias and maximal size.

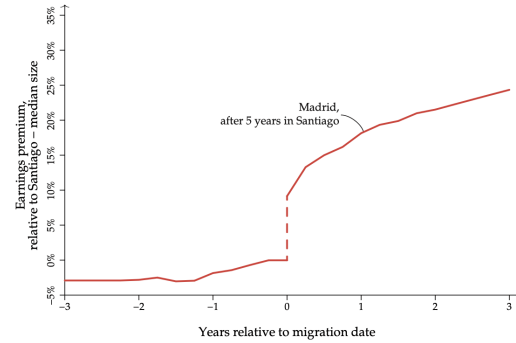


FIGURE 6
Non-parametric pre- and post-migration earnings profile relative to median-sized city

5.4 Table 3 and Figure 6: IV evidence and the no-Ashenfelter-dip check

Read:

- Table 3 shows that instrumenting city size barely changes the estimates: 0.0203 for the short-term premium and 0.0530 for the medium-term premium.
- The first-stage F -statistic is 25.2482, and the Hansen J -test does not reject the instruments.
- Figure 6 shows the nonparametric earnings profile around migration from Santiago to Madrid after five years in Santiago.
- There is no pre-migration dip in relative earnings.

Economic message. Two key threats are addressed here. The city-size results are not coming from endogenous city size, and the estimated migration gains are not just rebounds from temporary negative earnings shocks.

5.5 Table 4 and Figure 7: heterogeneity by worker ability

Read:

- Table 4 adds interactions between experience and worker fixed effects.
- The key interaction between experience in the two biggest cities and the worker fixed effect is positive and significant (0.0097), showing that more able workers gain more from big-city experience.
- With this specification, the short-term elasticity is 0.0243 and the medium-term elasticity is 0.0495.
- Figure 7 shows that after 10 years the Madrid-versus-Santiago premium is about 39% for a worker at the 75th percentile of ability and about 33% for a worker at the 25th percentile.
- The corresponding Sevilla-versus-Santiago gap is about 14% for high-ability workers and 12% for low-ability workers.

Dependent variable	(1) Log earnings	(2) Initial premium (city indicator coefficients column (1))	(3) Medium-term premium (initial + 7.7 years local experience)
Log city size		0.0243 (0.0061)***	0.0495 (0.0108)***
City indicators	Yes		
Worker fixed effects	Yes		
Experience first to second biggest cities	0.0293 (0.0022)***		
Experience first to second biggest cities × experience	-0.0007 (0.0001)***		
Experience third to fifth biggest cities	0.0143 (0.0043)***		
Experience third to fifth biggest cities × experience	-0.0006 (0.0003)**		
Experience	0.0979 (0.0007)***		
Experience ²	-0.0009 (0.0000)***		
Experience first to second biggest × worker fixed effect	0.0097 (0.0030)***		
Experience first to second biggest × experience	-0.0001 (0.0001)		
Experience third to fifth biggest cities × worker fixed effect	0.0042 (0.0045)		
Experience third to fifth biggest cities × experience	-0.0001 (0.0002)		
Experience × worker fixed effect	0.0632 (0.0034)***		
Experience ² × worker fixed effect	-0.0021 (0.0001)***		
Experience first to second biggest × now in five biggest	-0.0034 (0.0022)		
Experience first to second biggest × experience	0.0001 (0.0001)		
Experience third to fifth biggest × now in five biggest	-0.0029 (0.0003)		
Experience third to fifth biggest × experience	0.0003 (0.0003)		
Experience outside five biggest × now in five biggest	0.0022 (0.0024)		
Experience outside five biggest × experience	0.0000 (0.0001)		
Observations	6,263,446	76	76
R ²	0.1228	0.1352	0.3439

Notes: All regressions include a constant term. Column (1) also includes firm tenure and its square, occupation indicators, month-year indicators, two-digit sector indicators, and contract-type indicators. Coefficients in column (1) are reported with bootstrapped standard errors in parenthesis which are clustered by worker (achieving convergence of coefficients and mean squared error of the estimation in each of the 100 bootstrap iterations). Coefficients in columns (2) and (3) are reported with robust standard errors in parenthesis. ***, **, and * indicate significance at the 1, 5, and 10% levels. The R² reported in column (1) is within workers. Worker values of experience and tenure are calculated on the basis of actual days worked and expressed in years. City medium-term premium calculated for workers' average experience in one city (7.72 years).

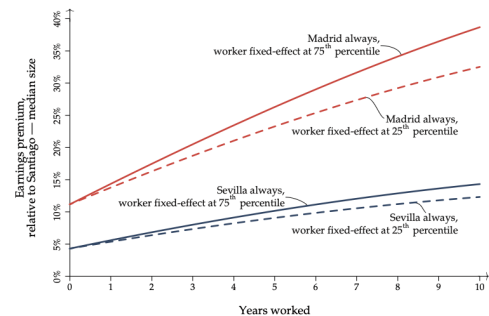


FIGURE 7
Earnings profiles relative to median-sized city, high- and low-ability worker

Economic message. Larger cities do not simply raise everyone's earnings equally. They amplify the payoff to ability by making experience more valuable for workers who can learn the most from it.

5.6 Table 5, Figure 8, and Table 6: what sorting really looks like

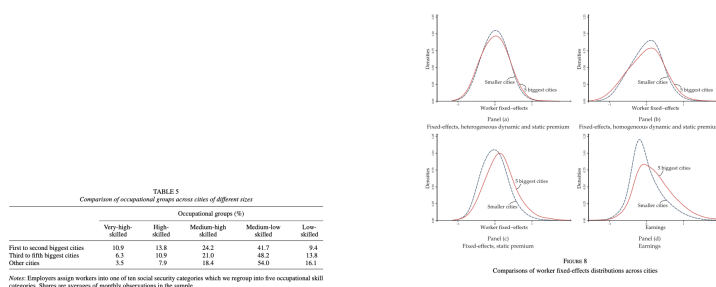


FIGURE 8
Comparisons of worker fixed-effects distributions across cities

Worker fixed-effects estimation	Shift (A)	Dilation (B)	Mean square quantile diff.	R ²	Obs.
---------------------------------	-----------	--------------	----------------------------	----------------	------

Notes: The table applies the methodology of [Cattaneo et al. \(2012\)](#) to approximate the distribution of worker fixed effects in the five biggest cities, F_{5big} , by taking the distribution of worker fixed effects in smaller cities, F_{small} , dilating it by a constant A , and dilating it by a factor D . A and D are estimated to minimize the mean quantile difference between the actual big city distribution F_{5big} and the dilated small city distribution $F_{small}(A, D)$. $R^2 = 1 - \text{MSE}(F_{5big}) / \text{MSE}(F_{small})$ is the total mean quantile difference between F_{5big} and $F_{small}(A, D)$. Coefficients are reported with bootstrapped standard errors in parenthesis (re-estimating worker fixed effects in each of the 100 bootstrap iterations). ***, **, and * indicate significance at the 1, 5, and 10% levels.

Read:

- Table 5 shows strong sorting on observables: very-high-skilled occupations account for 10.9% of jobs in the two biggest cities, 6.3% in cities ranked third to fifth, and 3.5% elsewhere.
- Figure 8, panel (a), plots worker fixed-effect distributions under the full heterogeneous dynamic model and shows that distributions in the five biggest cities and in smaller cities are very similar.
- Figure 8, panels (b) and (c), show that once dynamic heterogeneity is ignored, the distribution in bigger cities appears to have greater variance and then a higher mean as well.
- Figure 8, panel (d), shows that earnings themselves are both higher and more dispersed in bigger

cities.

- Table 6 formalizes this. Under the full model, the shift parameter comparing worker fixed-effect distributions across big and small cities is 0.0009 and statistically indistinguishable from zero, while the dilation parameter is 1.0854.
- Under a purely static specification, the shift parameter rises to 0.1571 and the dilation parameter to 1.1670; for earnings, the corresponding figures are 0.2210 and 1.2153.

Economic message. Big cities are full of more educated workers, but within broad skill groups they do not seem to contain much more innate unobserved ability. The larger mean and variance of earnings in big cities are mostly outcomes of working there, not just selection into them.

6 Short summary

- The paper studies why workers earn more in larger cities and asks how much is static, how much is sorting, and how much is learning.
- Using Spanish administrative data, it tracks complete work histories and allows the value of experience to depend on where it was acquired and where it is used.
- Standard pooled OLS gives a city-size elasticity of 0.0455; standard worker fixed effects lower it to 0.0241.
- Once dynamic learning is added, the short-term static elasticity is 0.0223 but the medium-term elasticity is 0.0510.
- Experience acquired in big cities is more valuable and is largely portable after workers move away.
- Big-city learning is stronger for high-ability workers.
- Sorting on observable skill categories is important, but within those broad categories there is little sorting on unobserved ability.
- The paper's big conclusion is that cities raise earnings not mainly because they attract inherently better workers, but because they teach workers more.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the earnings premium of bigger cities has two large pieces: an immediate static premium from currently working there and a dynamic premium from learning by working there. The dynamic premium is highly portable and becomes part of workers' human capital. Once this is taken into account, the case for large sorting on unobserved ability becomes much weaker.

7.2 Economic intuition

Larger cities place workers in denser, more productive environments. Some of the advantage is immediate: firms in bigger cities pay more because current production is more valuable there. But a large part is dynamic: workers in those cities accumulate more valuable experience. Since high-ability workers benefit even more from this process, larger cities end up with both higher average earnings and greater earnings dispersion.

7.3 Takeaway

The right way to think about the city-size wage premium is not “big cities attract better workers, so wages are higher.” The paper’s message is closer to: *big cities make workers better*. That is why the medium-term premium is so large, why it survives after workers move away, and why standard fixed-effects estimates understate the importance of learning.